

InvoiceDoc2

A full-stack Invoice Management System built with React, Express, and PostgreSQL.

Part 2

Installation

Required software & downloads

Install these before running the project:

01. Node.js (v20.19+ or 22.12+)

(Run server and client)

- Install LTS for Windows (.msi) with default options
- **Ensure "Add to PATH" is enabled**

<https://nodejs.org/>

02. Docker Desktop

(Run PostgreSQL)

- Install with default options
- Use Linux containers (not Windows containers)

<https://www.docker.com/products/docker-desktop/>

Required software & downloads

```
node -v
```

```
npm -v
```

```
docker --version
```

```
docker compose version
```

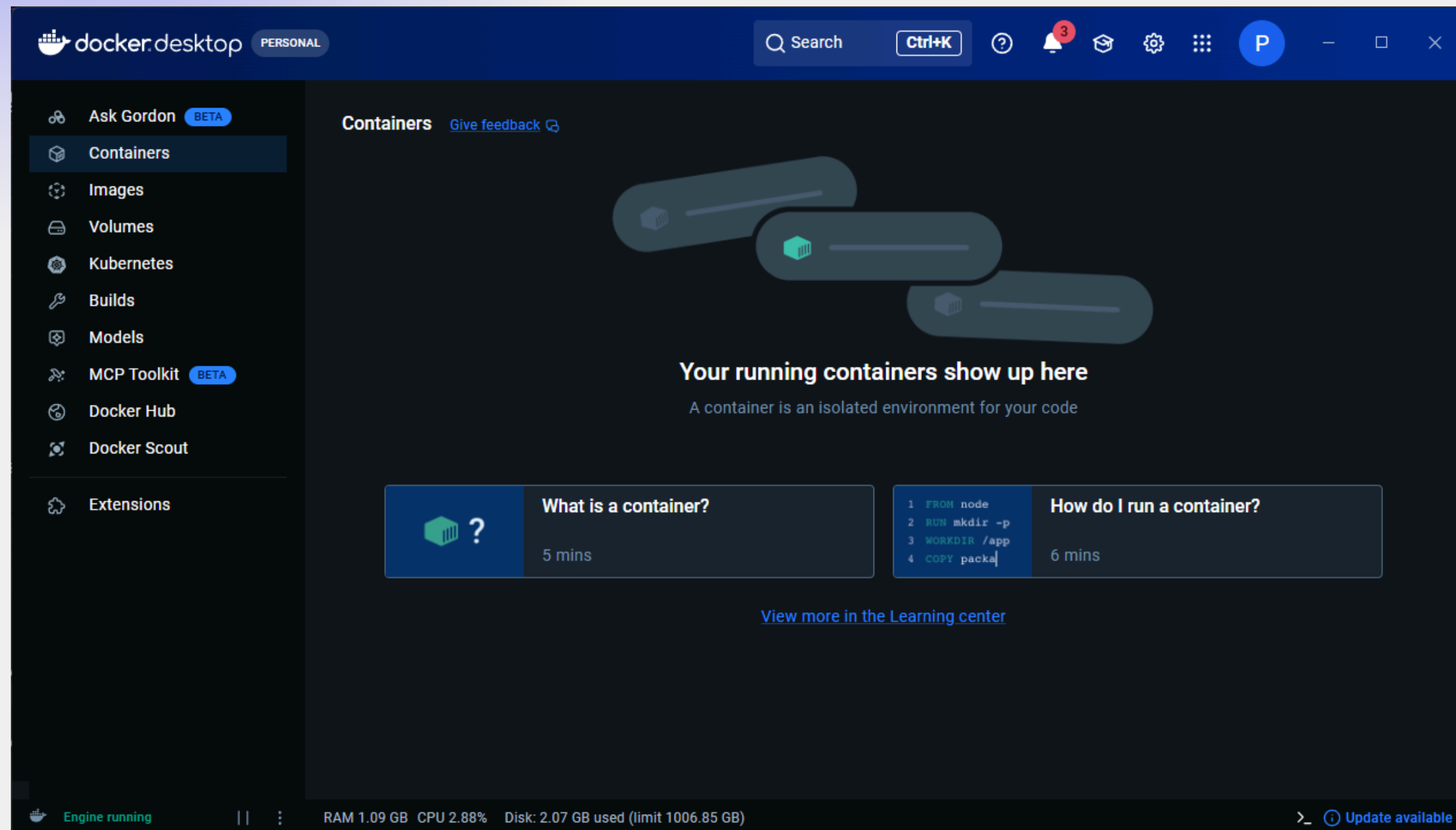
Check after install

```
PS E:\Work\InvoiceDoc2-main> node -v
v24.13.1
PS E:\Work\InvoiceDoc2-main> npm -v
11.8.0
PS E:\Work\InvoiceDoc2-main> docker --version
Docker version 29.2.1, build a5c7197
PS E:\Work\InvoiceDoc2-main> docker compose version
Docker Compose version v5.0.2
```

You should see the version numbers

Start The Project

Step 1: Start Docker Desktop



Open Docker Desktop and wait until it is fully running

Start The Project

Step 2: Start DB from repo root

Open terminal in project folder and run this command

`npm run docker:db:start`

```
PS E:\Work\InvoiceDoc2-main> npm run docker:db:start

> invoicedoc2@2.0.0 docker:db:start
> node scripts/docker-db-start.js

🚀 Starting InvoiceDoc2 Database...
❌ Docker is not running. Please start Docker Desktop first.
```

If you see warning like this, open the docker application first

```
PS E:\Work\InvoiceDoc2-main> npm run docker:db:start

> invoicedoc2@2.0.0 docker:db:start
> node scripts/docker-db-start.js

🚀 Starting InvoiceDoc2 Database...
🔧 Starting database and Adminer...
[+] up 36/36
✔Image adminer                               Pulled          19.8s
✔Image postgres:17-alpine3.22                Pulled          19.1s
✔Network database_compose_network             Created         0.1s
✔Volume database_pgdata                      Created         0.0s
✔Container database-pgdatabase-1              Created         0.5s
✔Container database-adminer-1                 Created         0.1s
⌚ Waiting for database...
ERROR: Input redirection is not supported, exiting the process immediately.
Error: Command failed: timeout /t 5 /nobreak > nul
ERROR: Input redirection is not supported, exiting the process immediately.

    at genericNodeError (node:internal/errors:985:15)
    at wrappedFn (node:internal/errors:539:14)
    at checkExecSyncError (node:child_process:925:11)
    at execSync (node:child_process:997:15)
    at execShell (E:\Work\InvoiceDoc2-main\scripts\run-safe.js:12:10)
    at main (E:\Work\InvoiceDoc2-main\scripts\docker-db-start.js:63:14)
    at process.processTicksAndRejections (node:internal/process/task_queues:103:5) {
  status: 1,
  signal: null,
  output: [
    null,
    <Buffer >,
    <Buffer 45 52 52 4f 52 3a 20 49 6e 70 75 74 20 72 65 64 69 72 65 63 74 69 6f 6e 20 69 73 20 6e 6f 74 20 73 75 70 70
6f 72 74 65 64 2c 20 65 78 69 74 69 6e 67 ... 27 more bytes>
  ],
  pid: 31712,
  stdout: <Buffer >,
  stderr: <Buffer 45 52 52 4f 52 3a 20 49 6e 70 75 74 20 72 65 64 69 72 65 63 74 69 6f 6e 20 69 73 20 6e 6f 74 20 73 75
70 70 6f 72 74 65 64 2c 20 65 78 69 74 69 6e 67 ... 27 more bytes>
}
PS E:\Work\InvoiceDoc2-main> |
```

Success output should look like this

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last st	Actions
☐	database	-	-	-	0%	5 minur	☐ ⋮
☐	pgdatabase-1	35195d3daab1	postgres:17	15432:5432 ↗	0%	5 minur	☐ ⋮
☐	adminer-1	4d9748d1847d	adminer	8080:8080 ↗	0%	5 minur	☐ ⋮

You should see the database in docker containers

Start The Project

Step 3: (Optional) Start Adminer

You can go to adminer (web UI) by clicking port number in container or go to this url

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last st	Actions
☐	database	-	-	-	0%	5 minur	  
☐	pgdatabase-1	35195d3daab1	postgres:1	15432:5432 ↗	0%	5 minur	  
☐	adminer-1	4d9748d1847d	adminer	8080:8080 ↗	0%	5 minur	  

Option1: Click the port number

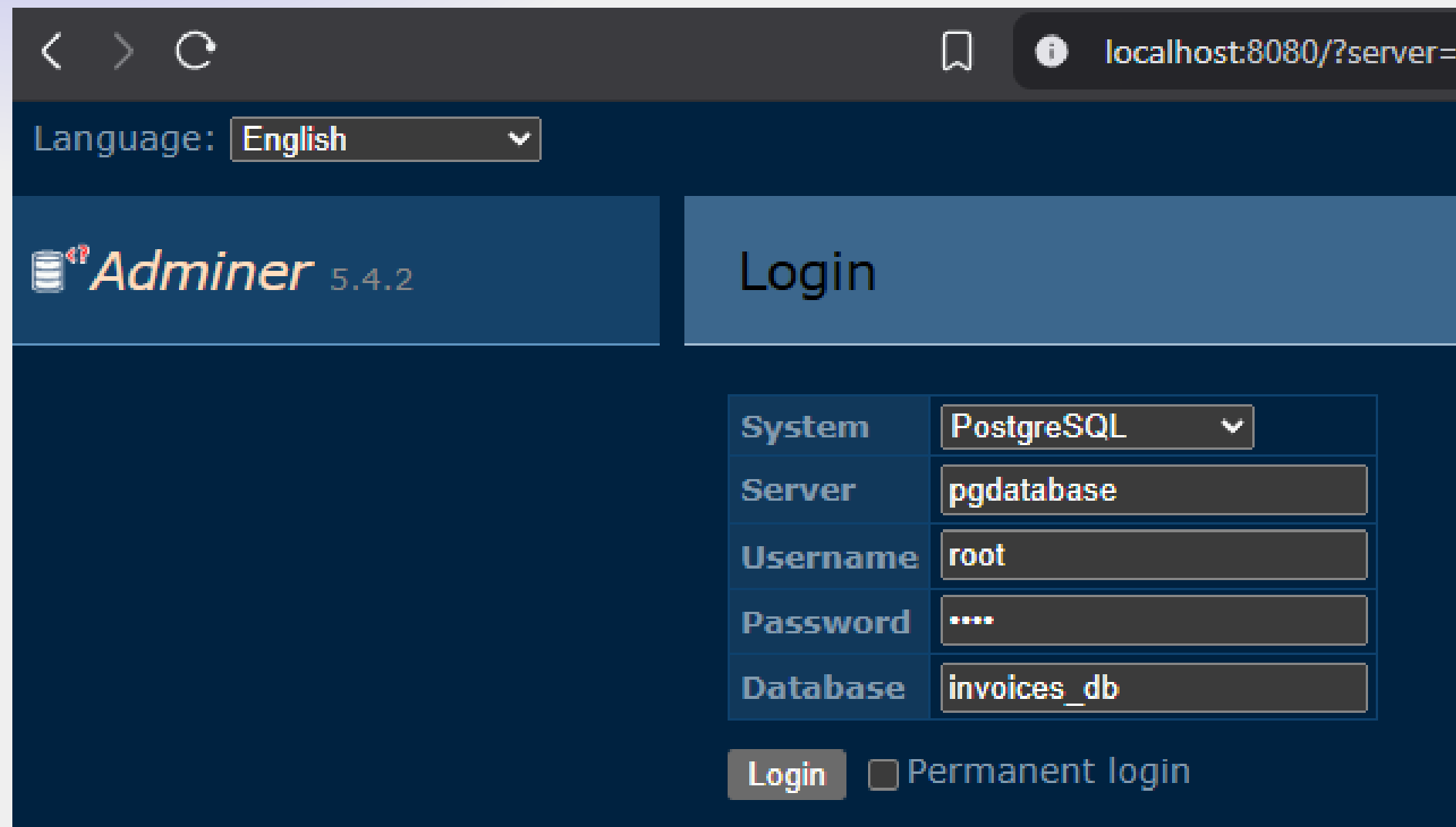
http://localhost:8080

Option2: Use this url

Start The Project

Step 3: (Optional) Start Adminer

Fill the login info like this



Language: English

Adminer 5.4.2

Login

System	PostgreSQL
Server	pgdatabase
Username	root
Password	****
Database	invoices_db

Login ☐ Permanent login

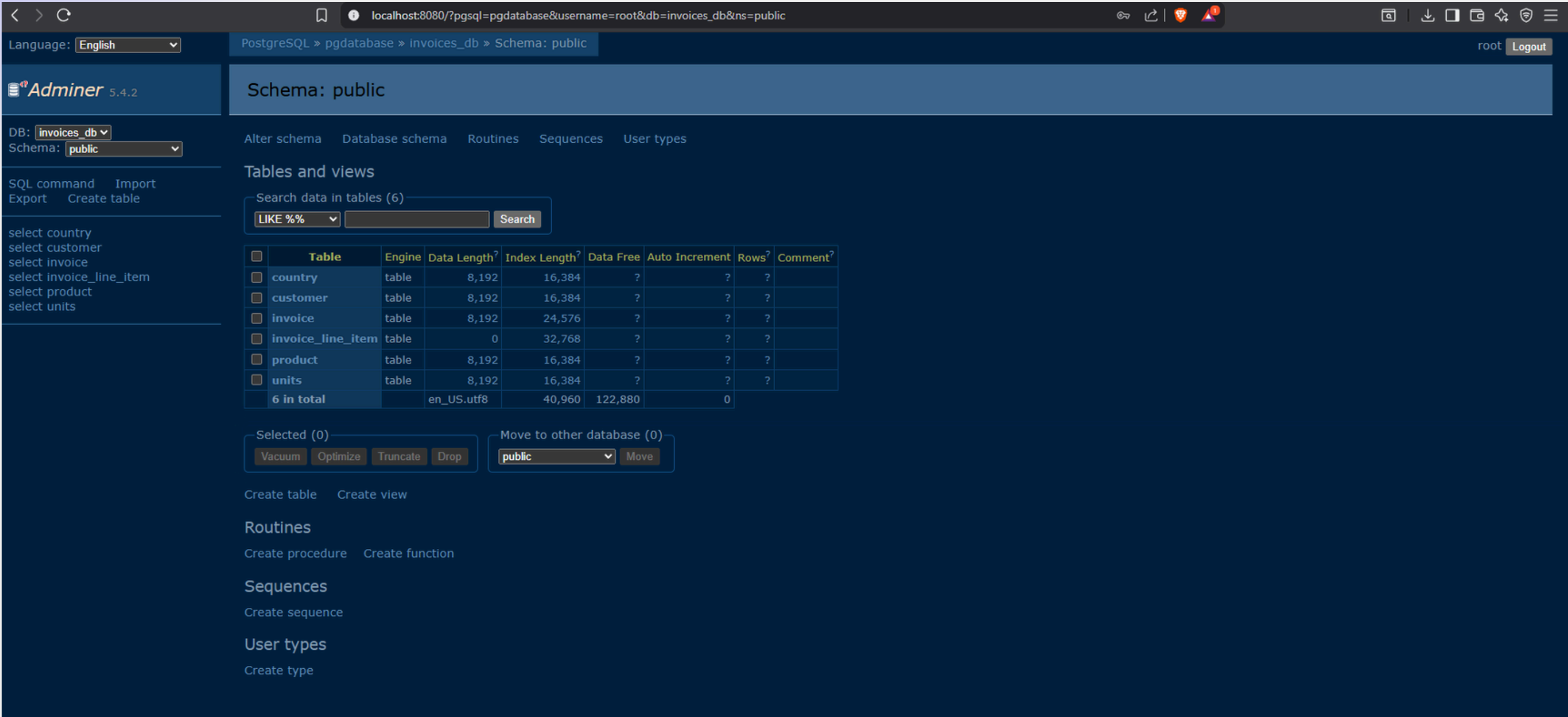
For database access, we use...

Host localhost : 15432
Database : invoices_db
User : root
Password : root

Start The Project

Step 3: (Optional) Start Adminer

After login, you should see this page



Start The Project

Step 4: Run API in terminal 2

Open new terminal and run these commands

```
cd server  
npm install  
npm audit  
npm audit fix
```

You should see this in the terminal

```
PS E:\Work\InvoiceDoc2-main> cd server  
PS E:\Work\InvoiceDoc2-main\server> npm install  
  
up to date, audited 115 packages in 910ms  
  
20 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities  
PS E:\Work\InvoiceDoc2-main\server> npm audit  
found 0 vulnerabilities  
PS E:\Work\InvoiceDoc2-main\server> npm audit fix  
  
up to date, audited 115 packages in 1s  
  
20 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities  
PS E:\Work\InvoiceDoc2-main\server>
```

Start The Project

Step 5: Create server/.env

This file stores local runtime settings (DB host/port/user/password, API port, keys).

Windows

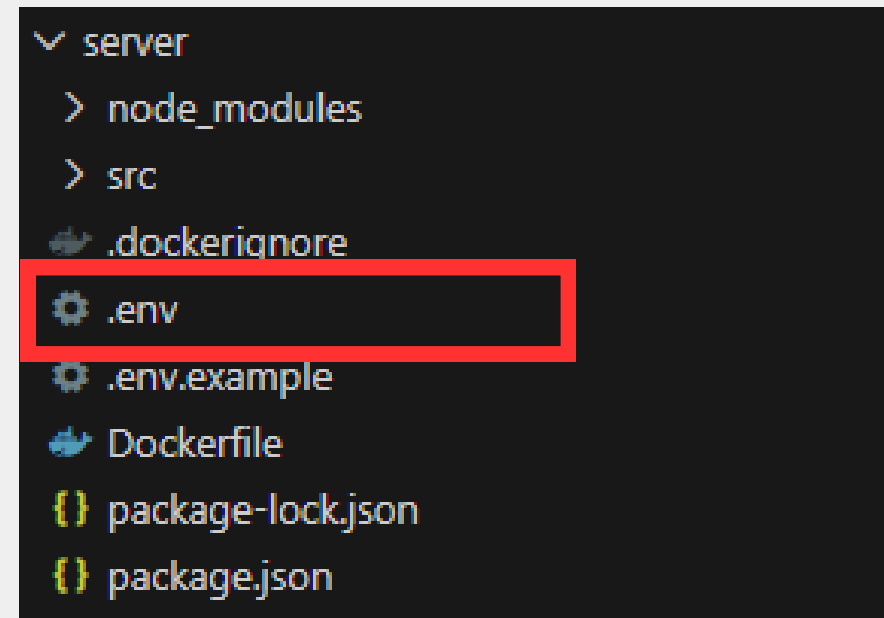
```
Copy-Item .env.example .env
```

Example on windows

```
PS E:\Work\InvoiceDoc2-main\server> Copy-Item .env.example .env
```

macOS/Linux

```
cp .env.example .env
```



```
server
├── node_modules
├── src
├── .dockerignore
├── .env
├── .env.example
├── Dockerfile
├── package-lock.json
├── package.json
```

You will see the .env file

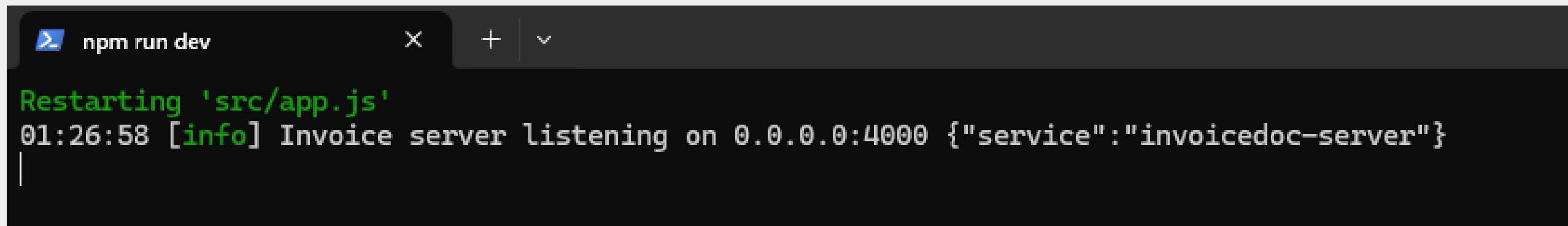
Start The Project

Step 6: Start API

Run this command in terminal 2

```
npm run dev
```

The result should look like this



```
npm run dev
Restarting 'src/app.js'
01:26:58 [info] Invoice server listening on 0.0.0.0:4000 {"service":"invoicedoc-server"}
|
```

Start The Project

Step 7: Run UI in terminal 3

Open new terminal and run this command

```
cd client  
npm install
```

The result should look like this

```
PS E:\Work\InvoiceDoc2-main> cd client  
PS E:\Work\InvoiceDoc2-main\client> npm install  
  
added 75 packages, and audited 76 packages in 3s  
  
11 packages are looking for funding  
  run `npm fund` for details  
  
1 high severity vulnerability  
  
To address all issues, run:  
  npm audit fix  
  
Run `npm audit` for details.  
PS E:\Work\InvoiceDoc2-main\client> |
```


Start The Project

Step 8: Create client/.env

Windows

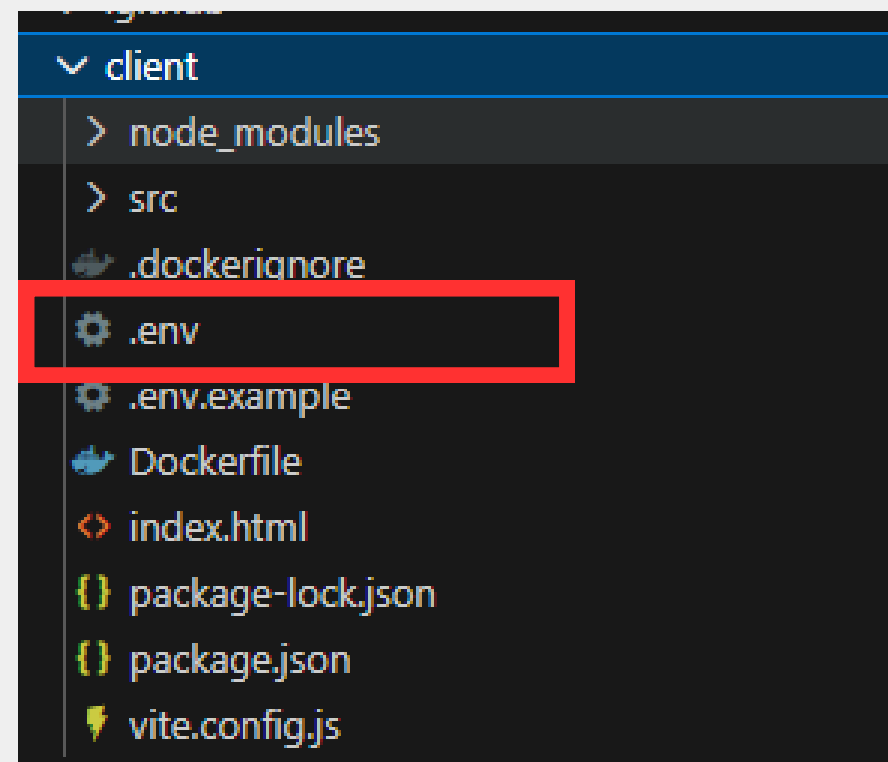
```
Copy-Item .env.example .env
```

macOS/Linux

```
cp .env.example .env
```

Example on windows

```
PS E:\Work\InvoiceDoc2-main\client> Copy-Item .env.example .env
```



You will see the .env file

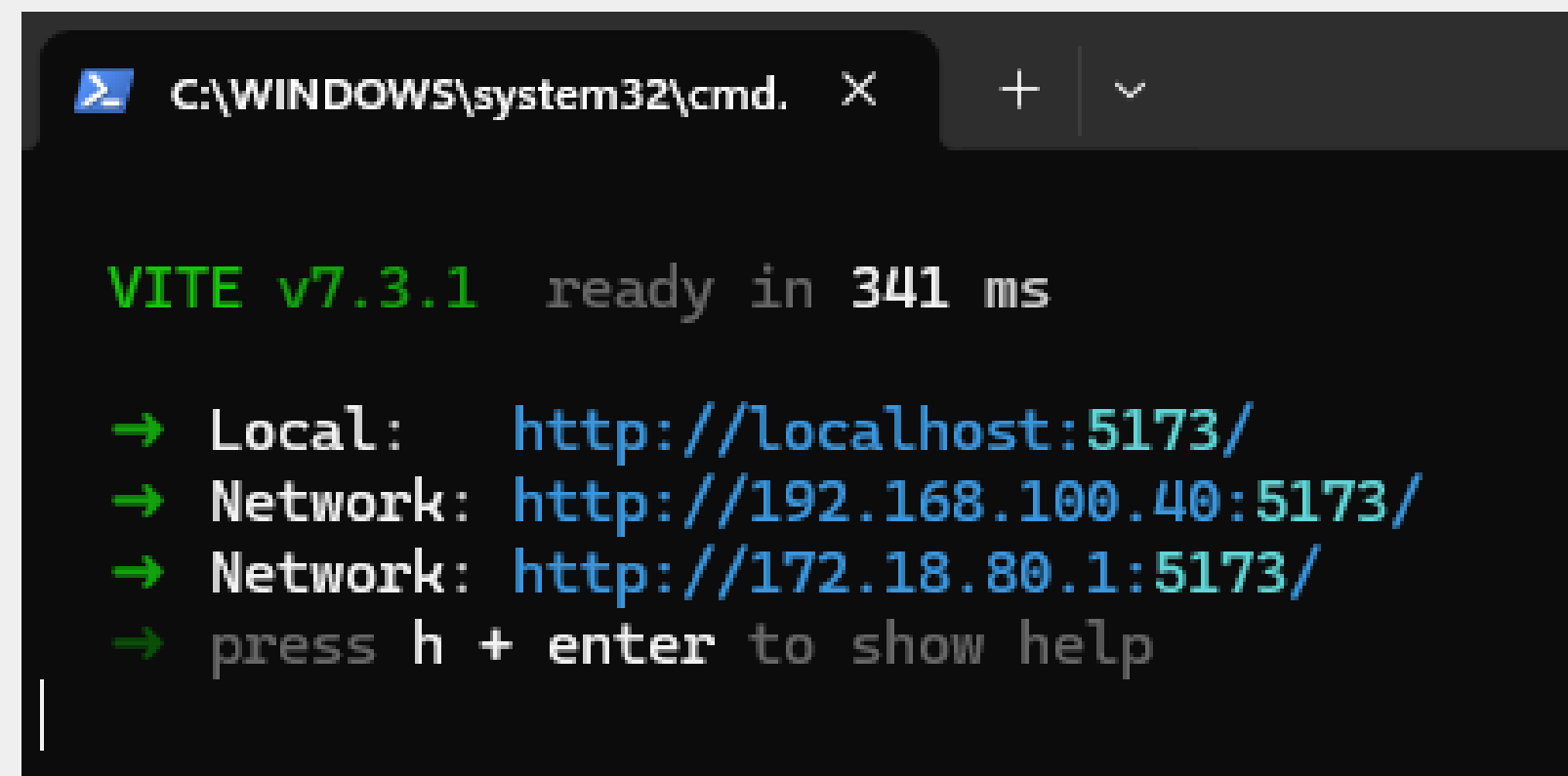
Start The Project

Step 9: Start UI

Run this command in terminal 3

```
npm run dev
```

The result should look like this

A screenshot of a Windows Command Prompt terminal window. The title bar shows the path 'C:\WINDOWS\system32\cmd.' and standard window controls. The terminal output displays 'VITE v7.3.1 ready in 341 ms' in green. Below this, four lines of information are shown with green arrows: 'Local: http://localhost:5173/', 'Network: http://192.168.100.40:5173/', 'Network: http://172.18.80.1:5173/', and 'press h + enter to show help'. A white cursor is visible at the bottom left of the terminal area.

```
> C:\WINDOWS\system32\cmd. X + v  
  
VITE v7.3.1 ready in 341 ms  
  
→ Local: http://localhost:5173/  
→ Network: http://192.168.100.40:5173/  
→ Network: http://172.18.80.1:5173/  
→ press h + enter to show help  
|
```

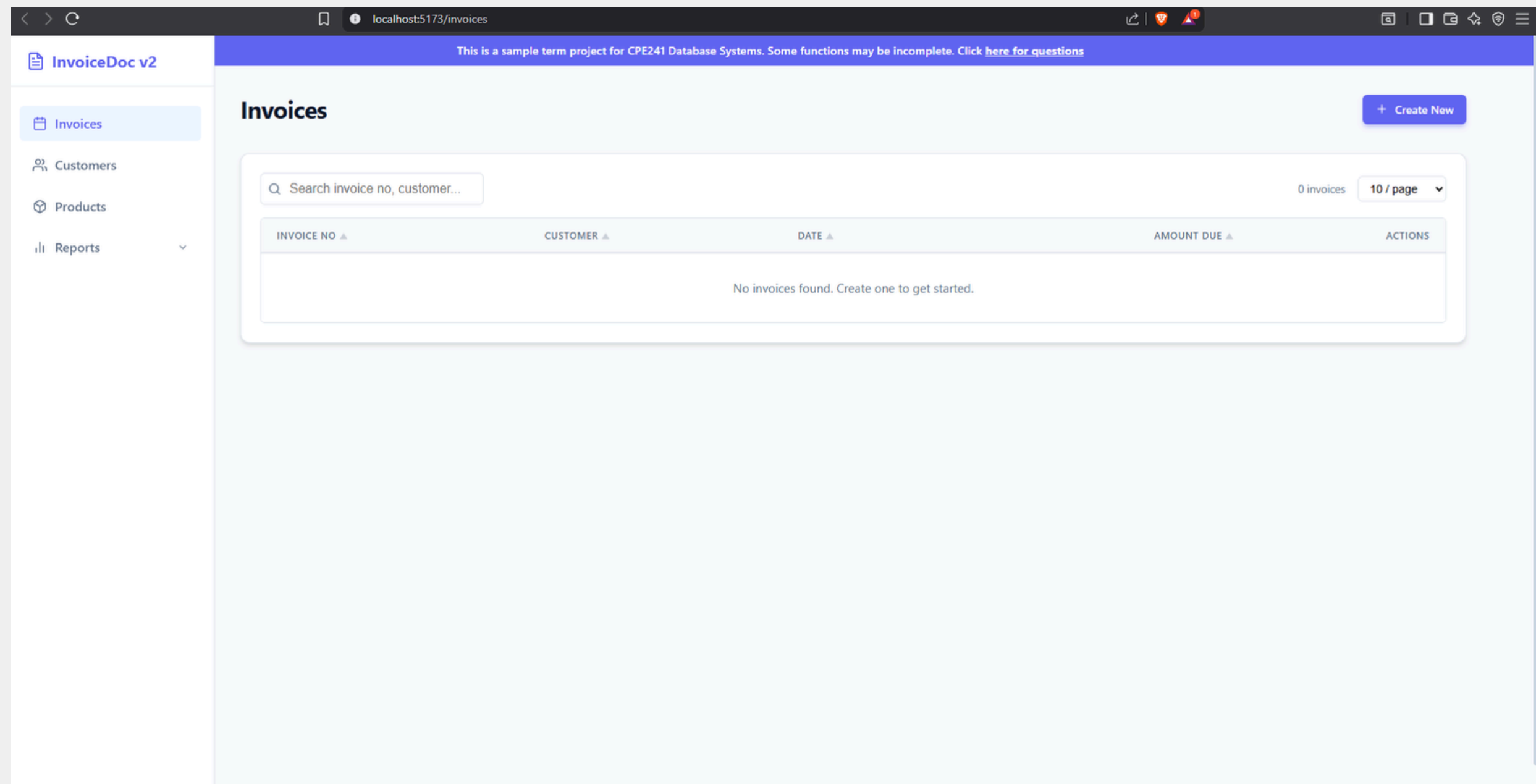

Start The Project

Step 10: Open and test app

UI Test

<http://localhost:5173>

UI Test Page



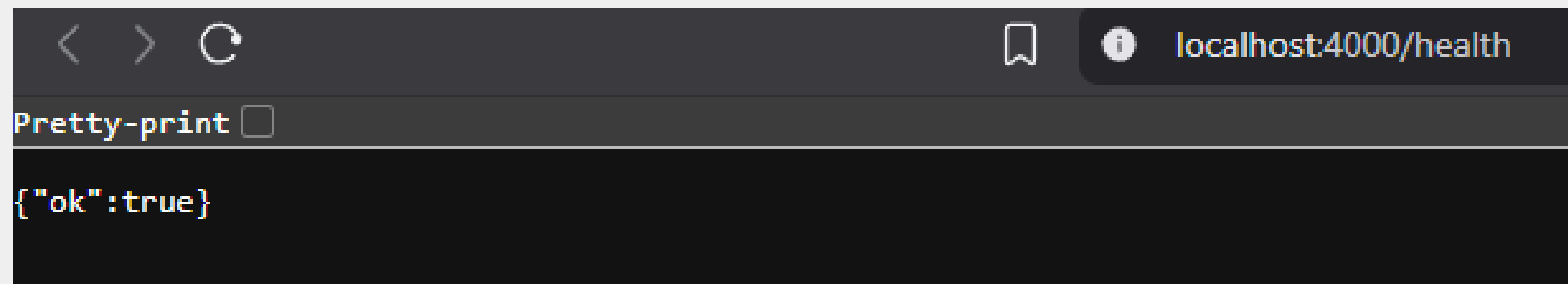
Start The Project

Step 10: Open and test app

API health

`http://localhost:4000/health`

Normal status output



Start The Project

Useful DB lifecycle commands (repo root)

Check status and row counts

npm run docker:db:check

```
PS E:\Work\InvoiceDoc2-main> npm run docker:db:check

> invoicedoc2@2.0.0 docker:db:check
> node scripts/docker-db-check.js

🔍 Checking InvoiceDoc2 Database...
✅ Container database-pgdatabase-1 is running

📊 Tables:
List of relations
Schema | Name | Type | Owner
-----+-----+-----+-----
public | country | table | root
public | customer | table | root
public | invoice | table | root
public | invoice_line_item | table | root
public | product | table | root
public | units | table | root
(6 rows)

📄 Counts:
customer: 0
product: 0
invoice: 0
invoice_line_item: 0

📍 localhost:15432 | invoices_db | root/root
```

View DB logs

npm run docker:db:logs

```
PS E:\Work\InvoiceDoc2-main> npm run docker:db:logs

> invoicedoc2@2.0.0 docker:db:logs
> docker-compose -f database/compose.yaml logs -f

pgdatabase-1 | The files belonging to this database system will be owned by user "postgres".
pgdatabase-1 | This user must also own the server process.
pgdatabase-1 |
pgdatabase-1 | The database cluster will be initialized with locale "en_US.utf8".
adminer-1 | [Sun Mar 1 17:35:06 2026] PHP 8.4.18 Development Server (http://[::]:8080) started
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36490 Accepted
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36490 [200]: GET /
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36490 Closing
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36502 Accepted
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36502 [200]: GET /?file=default.css&version=5.4.2
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36502 Closing
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36510 Accepted
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36514 Accepted
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36510 [200]: GET /?file=dark.css&version=5.4.2
pgdatabase-1 | The default database encoding has accordingly been set to "UTF8".
adminer-1 | [Sun Mar 1 17:40:53 2026] [::ffff:172.18.0.1]:36510 Closing
```

Stop the database

npm run docker:db:stop

```
PS E:\Work\InvoiceDoc2-main> npm run docker:db:stop

> invoicedoc2@2.0.0 docker:db:stop
> node scripts/docker-db-stop.js

🛑 Stopping InvoiceDoc2 Database...
[+] down 3/3
✅ Container database-adminer-1 Removed 0.4s
✅ Container database-pgdatabase-1 Removed 0.4s
✅ Network database_compose_network Removed 0.3s
✅ Database stopped!
```

Start The Project

Test the project working

Now you can try adding data to your project

InvoiceDoc v2

Invoices

Customers

Products

Reports

This is a sample term project for CPE241 Database Systems. Some functions may be incomplete. Click [here for questions](#)

Create Customer

← Back

Code

C001

☒ Auto

Name *

John Carton

Address Line 1

77 Lat Phrao Rd, Nonthaburi

Address Line 2

Address Line 2

Country *

Thailand

Credit Limit

10000

Create Customer

Continue your project

What to run next time you start your system

Step 1: Start Docker Desktop

Step 2: Terminal 1 (repo root)

```
# npm install # only needed first setup or when dependencies change  
npm run docker:db:start
```

Step 3: Terminal 2

```
cd server  
# npm install # only needed first setup or dependency changes# Copy-Item .env.example .env # done once on first setup  
npm run dev
```


Continue your project

What to run next time you start your system

Step 4: Terminal 3

```
cd client
```

```
# npm install
```

```
# only needed first setup or dependency changes # Copy-Item .env.example .env # done once on first setup
```

```
npm run dev
```

Step 5: Test API quickly

```
Invoke-RestMethod "http://localhost:4000/api/invoices?page=1&limit=5" | ConvertTo-Json -Depth 6
```

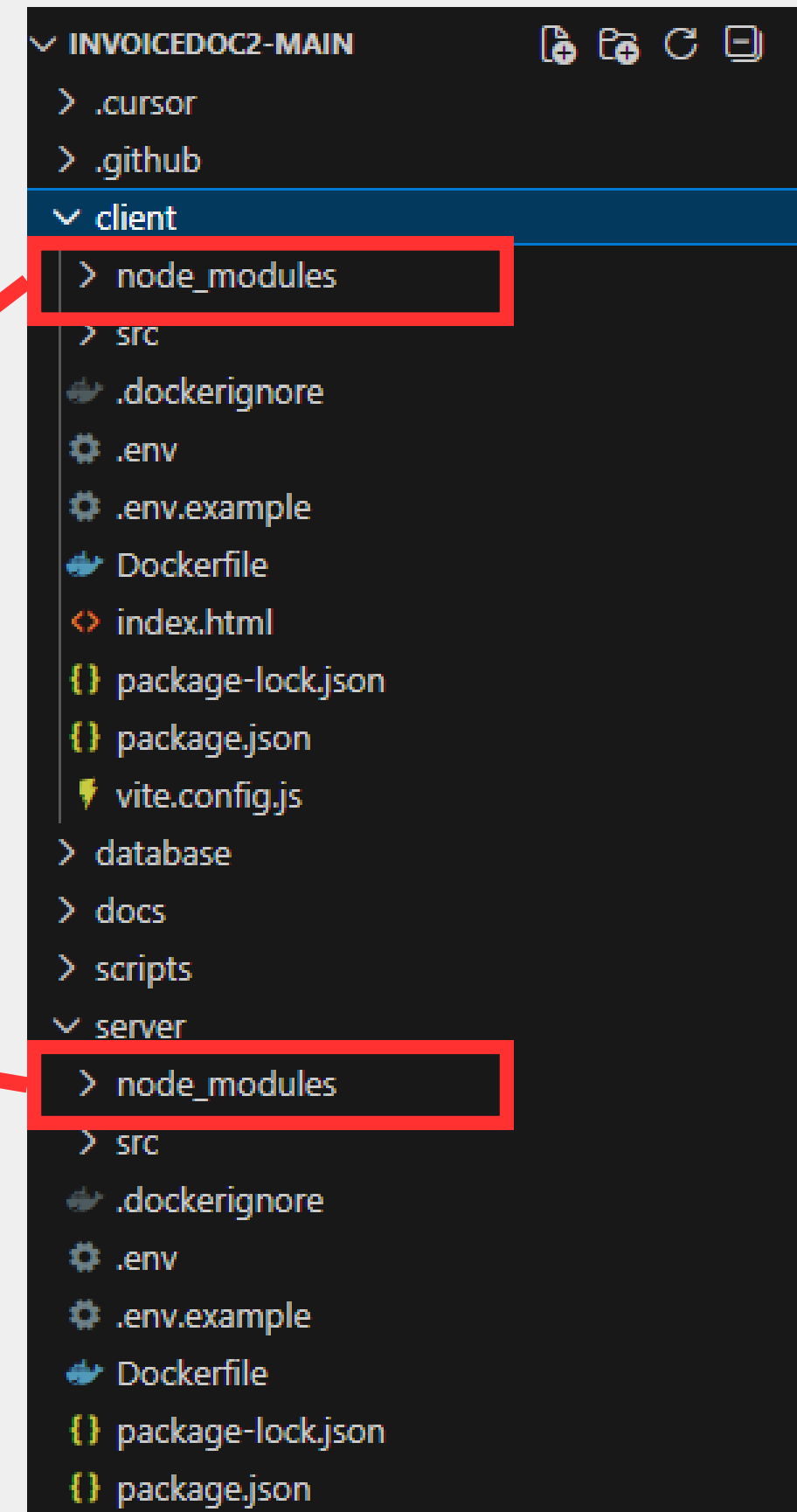
Example of API test

```
PS E:\Work\InvoiceDoc2-main> Invoke-RestMethod "http://localhost:4000/api/invoices?page=1&limit=5" | ConvertTo-Json -Depth 6
{
  "success": true,
  "data": [
  ],
  "error": null,
  "meta": {
    "total": 0,
    "page": 1,
    "limit": 5,
    "totalPages": 0
  }
}
```

Continue your project

Run npm install again only if:

- **First setup** on this machine
- Dependencies changed
- **node_modules** was deleted



Develop your project

Now it's time to get started and bring your project to life!